

Barbara Obayi

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PROFESSIONAL SUMMARY

Data scientist and machine learning engineer, MSc in Machine Learning and Deep Learning at the University of Strathclyde. Independently built FinWatch, a production machine learning system that scores bank customers for financial vulnerability under FCA Consumer Duty, with live ONS macro data at inference time. Led a 52-person cross-functional team to deliver a production computer vision system deployed on Android. Dissertation: a calibrated, auditable RAG system for versioned policy documents, with a trust calibration router. Three years of ML and analytical work on compliance-critical datasets for Singapore government clients.

EXPERIENCE

AI Research Engineer (MSc Dissertation)

May 2026 - Present

University of Strathclyde | Glasgow, UK

- Developing a question-answering AI system using large language models (LLMs) and retrieval-augmented generation (RAG) over a structured corpus of 100 versioned Strathclyde university policy documents.
- Implementing the pipeline in Python using LLaMA-3 8B, ChromaDB (vector database), sentence-transformers, and LangChain; benchmarking three retrieval strategies.
- Designing a calibration and evaluation framework that extends RAGAS with custom metrics including version correctness, citation fidelity, and abstention rate, alongside a trust routing layer.

Data Scientist and Product Owner

Sep 2024 - Jun 2025

Omdena (Kenya Chapter) | Remote

- Led a 52-person cross-functional team as Product Owner to deliver an AI-powered crop disease detection system for Kenyan smallholder farmers, from problem definition through live production deployment.
- Oversaw end-to-end computer vision pipeline: 17,939 labelled images, 4 crops, 20+ disease classes. Selected YOLO over ResNet and ViT based on accuracy-speed tradeoff; achieved 90-92% classification accuracy across crop categories.
- Deployed as an offline Android application using TensorFlow Lite. Defined evaluation benchmarks and quality thresholds before release; coordinated sprint planning across data, modelling, and app development groups.

Research and Data Analyst

Feb 2024 - Present

Meltwater (Z-Biz Solutions) | Remote, Singapore

- Delivered data science and analytical solutions for Singapore government clients including the Ministry of Manpower, Central Provident Fund, and Singapore Tourism Board, working with sensitive, compliance-critical datasets.
- Built automated Python and SQL data pipelines and Power BI dashboards; applied statistical modelling and analysis to generate actionable insights with material impact on senior policy decisions.
- Operated within strict data governance and compliance frameworks throughout client engagements, producing clear and auditable analytical outputs for non-technical executive audiences.

Data Analyst - Logistics and Operations

Mar 2022 - Oct 2022

Austel International | Nairobi, Kenya

- Automated data analysis and reporting pipelines across 10+ logistics partners using Python and Excel.

- Improved data accuracy and operational reporting efficiency, supporting procurement and delivery planning decisions across the business.

Project Manager Assistant

Aug 2021 - Mar 2022

Erdemann Property Limited | Nairobi, Kenya

- Improved on-time project delivery by 20% through structured data reporting, planning, and cross-functional coordination.

EDUCATION

MSc in Machine Learning and Deep Learning

2025 - 2026

University of Strathclyde | Glasgow, UK

Modules: Neural Networks and Deep Learning, Machine Learning for Data Analytics, Intelligent Sensing and Reasoning, Information Access and Mining, Big Data Technologies, Image and Video Processing

BEng in Civil Engineering

2018 - 2023

University of Shanghai for Science and Technology | Shanghai, China

- Awarded National First Prize by the China Business Statistics Society for a cross-cultural consumer behaviour study on smartphone purchase intentions, using SPSS and AMOS.

SELECTED PROJECTS

- [FinWatch](#): Production ML system for FCA Consumer Duty compliance. Scores 307,511 bank customers for financial vulnerability using XGBoost with live ONS macro data at inference time; PR-AUC 0.1877, 2.3x lift over baseline. Fairness gate enforced in code, SHAP explainability on every decision (GDPR Article 22), cost-sensitive threshold calibration. Full stack: FastAPI, Streamlit, MLflow, PSI drift monitoring, Docker, GitHub Actions CI/CD at 82% coverage.
- WorldQuant Applied Data Science (8 projects): Customer behaviour modelling, hypothesis testing, A/B testing, regression, classification, time-series forecasting.
- [Maji Ndogo Water Access Dashboard](#): Power BI dashboard for national and provincial water infrastructure decision-making across 5 provinces, using DAX measures, drill-through pages, and bookmark toggles to serve two distinct stakeholder audiences.
- [SQL Job Market Analysis](#): PostgreSQL and CTEs to identify the highest-paying remote data analyst roles and optimal skill combinations, mapping demand against average salary across real job postings.

SKILLS

ML and Deep Learning: Python, TensorFlow, Keras, PyTorch, scikit-learn, YOLO, ResNet, Vision Transformer (ViT), CNN, image classification, TensorFlow Lite

Data and Cloud: SQL, BigQuery (familiar), Google Cloud (familiar), PostgreSQL, NumPy, Pandas, automated pipelines, Git, GitHub, AWS, version control

Responsible AI: evaluation benchmark design, deployment quality thresholds, model documentation, compliance-aware data handling, governance frameworks

Experimentation: A/B testing, hypothesis testing, causal analysis, statistical modelling, KPI development, business impact measurement

Communication: stakeholder insight delivery, executive presentations, technical and non-technical audiences, Power BI, Tableau

Languages: English (native), Mandarin Chinese (professional), Swahili (native)